AS · Unit 1 · WPH11 Mechanics & Materials

Density, Upthrust & Viscous Drag

Why does a steel ship float when a steel bolt sinks? Why does a marble fall slowly through honey, and faster when the honey is warm? This topic is about how objects behave in fluids (liquids and gases). Three ideas do all the work: **density**, **upthrust** and **viscous drag**. Put them together and you can predict a ball's terminal velocity, which is exactly how Core Practical 2 measures the viscosity of a liquid.

Density

Density tells you how tightly mass is packed into a material: how many kilograms there are in each cubic metre.

Density

FORMULA SHEET

$$\rho = \frac{m}{V}$$

ρ: density / kg m⁻³ m: mass / kg V: volume / m³

A litre of water fills $1000\ \text{cm}^3 = 1.0 \times 10^{-3}\ \text{m}^3$ and has a mass of 1.0 kg, so its density is $1.0/1.0 \times 10^{-3} = 1000\ \text{kg m}^{-3}$. Density belongs to the **material**, not to the object: saw a steel bar in half and the mass and the volume both halve, so each half still has $\rho = 7800\ \text{kg m}^{-3}$.

DEFINITION

Density is mass per unit volume, $\rho = m/V$. Its SI unit is kg m⁻³.

Material	Density / kg m⁻³	Material	Density / kg m⁻³
air (20 °C)	1.2	sea water	1030
cork	≈ 240	glycerol	1260
pine wood	≈ 500	aluminium	2700
ice	917	steel	≈ 7800
vegetable oil	≈ 920	mercury	13 600
fresh water	1000	gold	19 300

COMMON MISTAKE

Volume units are the classic trap. $1\ \text{cm} = 10^{-2}\ \text{m}$, so $1\ \text{cm}^3 = (10^{-2})^3\ \text{m}^3 = 10^{-6}\ \text{m}^3$, and $1\ \text{mm}^3 = 10^{-9}\ \text{m}^3$. A handy check: $1\ \text{g cm}^{-3} = 1000\ \text{kg m}^{-3}$, so water is 1.00 g cm⁻³.

Measuring density

- **Regular solid:** mass on a balance; measure the sides with a ruler, vernier calipers or micrometer and calculate the volume.
- **Irregular solid:** mass on a balance; find the volume by **displacement**. Lower it into a measuring cylinder of water and read the rise in level, or use a displacement (Eureka) can and collect the water that overflows.
- **Liquid:** weigh an empty measuring cylinder, pour in a measured volume and weigh it again.

WORKED EXAMPLE

A crown has a mass of 1.20 kg. When it is lowered into a full displacement can, 75 cm³ of water overflows. The density of gold is 19 300 kg m⁻³. Is the crown pure gold?

Show solution

The crown displaces its own volume: $V = 75\ \text{cm}^3 = 75 \times 10^{-6}\ \text{m}^3$.

$$\rho = \frac{m}{V} = \frac{1.20}{75 \times 10^{-6}} = 1.6 \times 10^{4}\ \text{kg m}^{-3}$$

That is well below 19 300 kg m⁻³, so the crown is not pure gold; it has been mixed with a less dense metal. A pure gold crown of the same mass would displace only $1.20/19\,300 = 6.2 \times 10^{-5}\ \text{m}^3 = 62\ \text{cm}^3$. (This is the problem Archimedes is said to have solved in his bath.)

Upthrust

Push an empty plastic bottle under water and you can feel it shove back up. Every object in a fluid feels an upward force from that fluid. This force is called **upthrust**, and it obeys a beautifully simple rule.

Archimedes' principle

$$\text{upthrust} = \text{weight of fluid displaced} \qquad U = \rho_\text{f}\, V_\text{disp}\, g$$

U: upthrust / N ρ_f: density of the fluid / kg m^{-3}

V_disp: volume of fluid pushed aside (the submerged volume) / m^3 $g = 9.81\ \text{N kg}^{-1}$

The second form is just the first one written out: the displaced fluid has mass $\rho_\text{f} V_\text{disp}$ (from $\rho = m/V$), and its weight is that mass times g.

Why the rule works

Here is a lovely thought experiment. Take the object out and imagine the hole it leaves is filled with more of the fluid, the same shape as the object. That blob of fluid just sits there: it doesn't sink or rise, because it's simply part of the fluid. So the surrounding fluid must be pushing up on the blob with a force equal to the blob's **weight**.

Now put the object back. The surrounding fluid can't tell what is inside that surface: it pushes on the object in exactly the same way it pushed on the blob. So the upthrust on the object equals the weight of the fluid it has displaced, whatever the object is made of.

KEY IDEA

What upthrust does and doesn't depend on

- It depends on the **volume submerged** and the **density of the fluid**. Salt water gives more upthrust than fresh water.
- It does **not** depend on the object's mass or material. A steel ball and a plastic ball of the same size, both fully under water, feel the same upthrust.
- Once an object is fully submerged, pushing it deeper doesn't change the upthrust (liquids are almost incompressible, so the displaced volume stays the same).

DEFINITION

Upthrust is the upward force that a fluid exerts on an object that is wholly or partly immersed in it. It is equal to the weight of the fluid displaced.

Weighing things in water

Hang an object from a newton meter and lower it into water: the reading falls. The newton meter now only has to supply the part of the weight that the upthrust doesn't, so the reading is $W - U$, sometimes called the **apparent weight**. A displacement can lets you check Archimedes' principle directly: the water that overflows is the water displaced, so you can weigh it.

WORKED EXAMPLE

A stone hangs from a newton meter. The reading is 5.0 N in air and 3.2 N when the stone is completely submerged in water (density 1000 kg m^{-3}). Calculate (a) the upthrust, (b) the volume of the stone and (c) its density.

Show solution

(a) The newton meter reads $W - U$, so

$$U = 5.0 - 3.2 = 1.8 \text{ N}$$

(b) The upthrust is the weight of water displaced, and the stone displaces its own volume:

$$U = \rho_{\text{w}} V g \;\Rightarrow\; V = \frac{U}{\rho_{\text{w}} g} = \frac{1.8}{1000 \times 9.81} = 1.83 \times 10^{-4} \text{ m}^3$$

(c) Mass from the weight in air, then density:

$$m = \frac{W}{g} = \frac{5.0}{9.81} = 0.510 \text{ kg}, \qquad \rho = \frac{m}{V} = \frac{0.510}{1.83 \times 10^{-4}} = 2.8 \times 10^{3} \text{ kg m}^{-3}$$

Shortcut worth noticing: $W/U = \rho_{\text{stone}} V g / \rho_{\text{w}} V g = \rho_{\text{stone}} / \rho_{\text{w}}$, so $\rho_{\text{stone}} = 1000 \times 5.0/1.8 = 2800 \text{ kg m}^{-3}$ in one line.

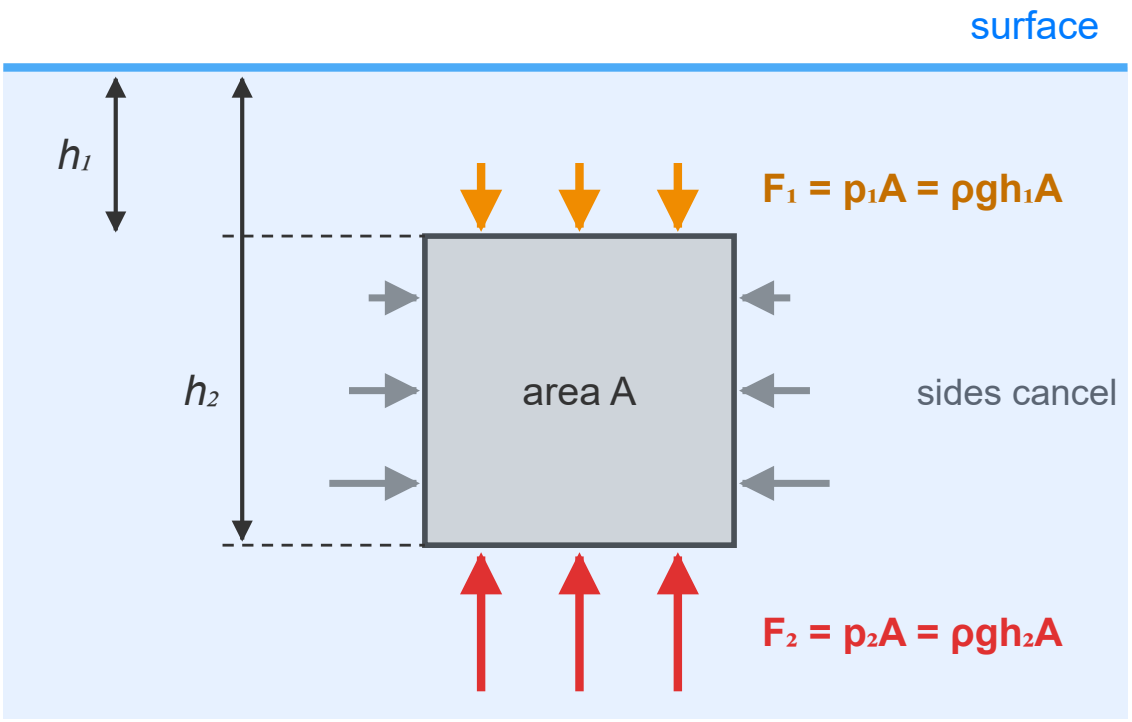
EXAM TIP

Air is a fluid too, so everything in air feels a small upthrust. A party balloon 30 cm across displaces about 0.014 m³ of air, giving an upthrust of $1.2 \times 0.014 \times 9.81 \approx 0.17$ N. Fill it with helium (much less dense than air) and that is more than its weight, so it rises. For dense objects like a steel ball, upthrust in air is tiny and exam questions let you ignore it, but in a **liquid** you must include it.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Where upthrust comes from: pressure increases with depth

The thought experiment tells you how big upthrust is. The underlying cause is that the pressure in a fluid increases with depth, so the fluid pushes harder on the bottom of an object than on its top. Hydrostatic pressure $p = \rho g h$ isn't in the IAL spec, but it makes the whole thing click.



The deeper bottom face is pushed up harder than the top face is pushed down. The sideways forces cancel in pairs.

Upthrust from the pressure difference

NOT PART OF THE EDEXCEL SPEC

1. The pressure at depth h in a fluid comes from the weight of the column of fluid above. A column of area A and height h has mass $\rho A h$, so its weight spread over the area A is

$$p = \frac{\rho A h g}{A} = \rho g h$$

Atmospheric pressure acts on every face as well, but it adds the same amount everywhere, so it cancels out of the difference below.

2. The top face (area A, depth h_1) is pushed down and the bottom face (depth h_2) is pushed up:

$$F_1 = \rho_\text{f} g h_1 A \qquad F_2 = \rho_\text{f} g h_2 A$$

3. The side forces cancel in pairs, so the resultant upward force is

$$U = F_2 - F_1 = \rho_\text{f} g A (h_2 - h_1)$$

4 $A(h_2 - h_1)$ is the volume of the block, V, so

$$U = \rho_{\mathrm{f}} V g$$

and $\rho_{\mathrm{f}} V$ is the mass of fluid that would fill that space:

$$\boxed{U = \text{weight of fluid displaced}}$$

Only the difference $h_2 - h_1$ appears, which is why a fully submerged object feels the same upthrust at any depth.

Floating and sinking

Take an object of volume V and push it completely under. Its weight is $W = \rho_{\mathrm{obj}} V g$ and the upthrust is $U = \rho_{\mathrm{f}} V g$. The V and g are the same in both, so it all comes down to comparing densities:

- $\rho_{\mathrm{obj}} > \rho_{\mathrm{f}}$: $W > U$, so it **sinks**. On the bottom, the floor supplies the missing force: $W = U + R$, where R is the normal contact force.
- $\rho_{\mathrm{obj}} < \rho_{\mathrm{f}}$: $U > W$, so it **rises** until part of it sticks out. As it rises, the submerged volume shrinks and so does U, until $U = W$ and it **floats**.
- $\rho_{\mathrm{obj}} = \rho_{\mathrm{f}}$: it stays wherever you put it (neutral buoyancy, like a fish or a submarine).

How much of a floating object is under the surface?

1 A floating object is in equilibrium, so the resultant force on it is zero:

$$U = W$$

2 The upthrust is the weight of the fluid displaced by the submerged volume V_{sub}. The weight uses the whole volume V:

$$\rho_{\mathrm{f}} V_{\mathrm{sub}}\, g = \rho_{\mathrm{obj}} V g$$

3 Cancel g and divide both sides by $\rho_{\mathrm{f}} V$:

$$\boxed{\frac{V_{\mathrm{sub}}}{V} = \frac{\rho_{\mathrm{obj}}}{\rho_{\mathrm{f}}}}$$

g cancels, so a boat floats at the same level on the Moon.

Ice (917 kg m⁻³) in sea water (1030 kg m⁻³) floats with $917/1030 = 0.89$ of its volume under the surface, which is why "the tip of the iceberg" is only about a tenth of it. A steel ship floats because what matters is its **average** density: the hull encloses a huge volume of air, so the ship's total mass divided by the volume of the hull is less than 1030 kg m⁻³. Load it with cargo and it sits lower in the water, displacing more water to balance the extra weight.

WORKED EXAMPLE

A rectangular block of wood measures 0.50 m × 0.20 m × 0.10 m. It floats in water (density 1000 kg m⁻³) with its largest face horizontal and 6.5 cm of its 10 cm thickness under the surface. (a) Calculate the density of the wood. (b) Calculate the smallest mass that could be placed on top of the block to push it completely under.

Show solution

(a) The cross-section is the same all the way down, so the fraction of the volume submerged is the fraction of the thickness submerged:

$$\frac{\rho_{\mathrm{wood}}}{\rho_{\mathrm{w}}} = \frac{6.5}{10} \;\Rightarrow\; \rho_{\mathrm{wood}} = 0.65 \times 1000 = 650 \ \mathrm{kg\ m^{-3}}$$

(b) Volume $V = 0.50 \times 0.20 \times 0.10 = 0.010 \ \mathrm{m^3}$. Fully submerged, the upthrust and the block's weight are

$$U = 1000 \times 0.010 \times 9.81 = 98.1 \ \mathrm{N}, \qquad W = 650 \times 0.010 \times 9.81 = 63.8 \ \mathrm{N}$$

The extra weight needed is $98.1 - 63.8 = 34.3 \ \mathrm{N}$, so the mass is

$$m = \frac{34.3}{9.81} = 3.5 \ \mathrm{kg}$$

Viscosity and Stokes' law

Viscosity is how "thick" a fluid is: its resistance to flowing. Honey has a high viscosity and water a low one. When a fluid flows, neighbouring layers slide past each other, and viscosity is like an internal friction between those layers. Its symbol is η (eta).

Now drop a ball into the fluid. The layer of fluid touching the ball is dragged along with it, the next layer out is dragged a little less, and so on: the ball has to keep shearing the fluid around it. That takes a force, so the fluid pulls back on the ball. This is **viscous drag**. For a small sphere moving slowly, George Stokes worked out exactly how big it is.

Stokes' law: viscous drag on a sphere

FORMULA SHEET

$$F = 6\pi\eta r v$$

F: viscous drag force / N η: viscosity of the fluid / Pa s r: radius of the sphere / m
v: speed of the sphere relative to the fluid / m s^{-1}

Why each factor is there

- η: a stickier fluid resists the shearing more.
- v: moving faster means the layers shear past each other faster. At low speeds the drag is simply proportional to the speed.
- r: a bigger sphere drags a bigger region of fluid along with it.
- 6π: this number comes from the geometry of a sphere. Working it out needs maths well beyond A level, so just use it.

Rearranging gives the unit of viscosity: $\eta = F/(6\pi r v)$ has units $\mathrm{N}/(\mathrm{m} \times \mathrm{m\ s^{-1}}) = \mathrm{N\ s\ m^{-2}}$. Since $1\ \mathrm{N\ m^{-2}} = 1\ \mathrm{Pa}$, that is the **pascal second, Pa s**, which in base units is kg m^{-1} s^{-1}.

KEY IDEA

When can you use Stokes' law?

Only for **small**, **spherical** objects moving at **low speeds** with **laminar flow** (no turbulence) around them. A raindrop in air or a ball bearing in thick oil is fine; a football, a car or a skydiver is not. And because viscosity is **temperature dependent**, the value of η you use must be the one for the temperature of the fluid.

Viscosity depends on temperature

For liquids, viscosity falls steeply as the temperature rises. Golden syrup pours easily after the tin has stood in hot water, and cold engine oil is so thick that engines wear fastest just after starting. Glycerol is about **five times** less

viscous at 40 °C than at 20 °C. So any viscosity value only means something if you also give the temperature, and in an experiment you must measure the temperature and keep it steady.

Fluid (at 20 °C)	Viscosity η / Pa s
air	1.8×10^{-5}
water	1.0×10^{-3}
olive oil	≈ 0.08
castor oil	≈ 1.0
glycerol	≈ 1.4
honey	roughly 2 to 10 (depends on the honey)

⚠ Beyond the spec · not part of the Edexcel IAL specification

Why hot liquids are runnier, but hot gases are stickier

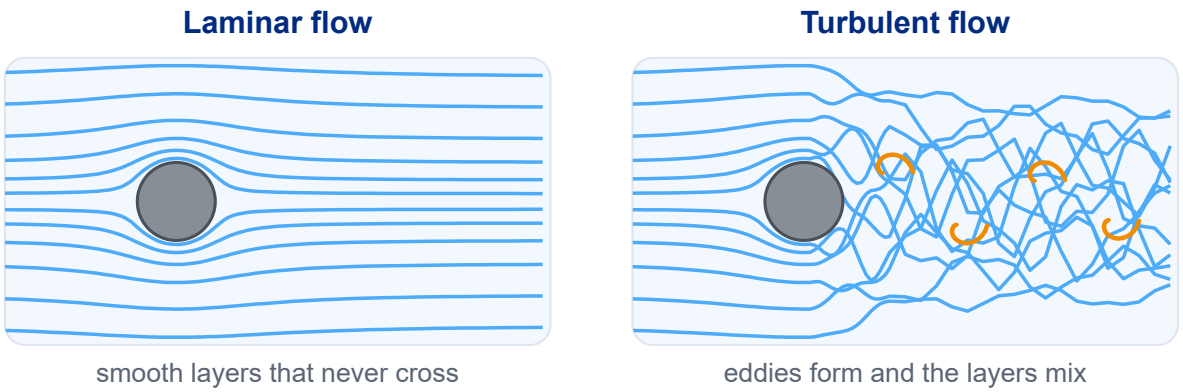
In a liquid the molecules are packed closely and attract their neighbours. For one layer to slide over another, molecules have to keep breaking free of the neighbours holding them. Heat the liquid and the molecules have more kinetic energy, so far more of them can escape at any moment: the viscosity drops roughly exponentially with temperature.

Gases do the opposite. Their molecules hardly attract each other at all; their viscosity comes from molecules flying between layers and carrying momentum with them, which drags the faster and slower layers towards the same speed. Hotter molecules move faster and carry more momentum across, so a gas becomes **more** viscous when heated (air goes from about 1.8×10^{-5} Pa s at 20 °C to about 2.2×10^{-5} Pa s at 100 °C).

Laminar and turbulent flow

To picture a flow we draw **streamlines**: lines that show the path a small piece of the fluid follows. Where the streamlines squeeze closer together, the fluid is moving faster (the same amount of fluid is getting through a narrower gap).

- In **laminar flow** the fluid moves in smooth layers that slide past each other without mixing. The streamlines are smooth and never cross, and the velocity at any point stays the same over time.
- In **turbulent flow** the fluid swirls into eddies and mixes. The flow pattern keeps changing, and the speed and direction at any point change erratically.



Fluid flowing from left to right past a sphere. Laminar flow (left): smooth, non-crossing streamlines, bunched up where the fluid speeds past the sides of the sphere. Turbulent flow (right): behind the sphere the flow breaks into eddies and the layers mix.

DEFINITION

Laminar flow: the fluid flows in layers with no mixing between them; streamlines do not cross and the velocity at each point is constant.

Turbulent flow: the fluid mixes in irregular eddies; the speed and direction of flow at each point change with time.

Flow round an object tends to be laminar when the object is **small**, moving **slowly** and the fluid is **viscous**. Speed it up and a wake of eddies forms behind it. Stokes' law assumes the smooth, layered flow, so it only works in the laminar case. Once there is turbulence the object also has to keep setting the fluid swirling, which costs extra energy, so the drag becomes bigger than $6\pi\eta r v$.

EXAM TIP

A favourite question: "Explain why Stokes' law cannot be used to find the drag on a skydiver." Give the conditions it breaks: a skydiver is not small, not spherical and moves fast, so the air flow around them is turbulent.

⚠ Beyond the spec · not part of the Edexcel IAL specification

The Reynolds number, and drag at high speed

Engineers predict whether a flow will be laminar or turbulent using one number, the **Reynolds number**, which compares the fluid's tendency to keep moving (its inertia) with the viscous forces trying to smooth it out:

$$Re = \frac{\rho_{\mathrm{f}}\, v\, d}{\eta}$$

where d is the diameter of the sphere. Stokes' law is accurate when Re is less than about 1. A 2 mm steel ball falling at 1 cm s^{-1} through glycerol has $Re = 1260 \times 0.01 \times 0.002 / 1.4 \approx 0.02$, comfortably laminar. Drop the same ball in water and Stokes' law predicts about 15 m s^{-1}, but the real ball falls at well under 1 m s^{-1} with Re above 1000 and a turbulent wake.

At high Reynolds numbers the drag is roughly proportional to v^2 instead of v: $F \approx \frac{1}{2} C_\mathrm{d} \rho_\mathrm{f} A v^2$, where A is the frontal area and C_d (the drag coefficient) depends on the shape. That is why cyclists and cars care so much about streamlining, and why doubling your speed on a bike roughly quadruples the air resistance.

Terminal velocity

Let's put the three ideas together. A small sphere of radius r and density ρ_s is released from rest in a liquid of density ρ_f and viscosity η. Three forces act on it:

- **Weight** $W = mg = \frac{4}{3}\pi r^3 \rho_\mathrm{s} g$, downwards.
- **Upthrust** $U = \frac{4}{3}\pi r^3 \rho_\mathrm{f} g$, upwards (the weight of liquid displaced).
- **Viscous drag** $F = 6\pi\eta r v$, upwards (always opposite to the velocity).

1. At the instant of release $v = 0$, so there is no drag. The resultant force is $W - U$ and the ball accelerates downwards (with less than g, because of the upthrust).
2. As the ball speeds up, the drag grows in proportion to v. The resultant force $W - U - F$ shrinks, so the acceleration gets smaller and smaller.
3. Eventually F has grown so that $W = U + F$. The resultant force is zero, so (by Newton's first law) the ball carries on at a constant velocity: its **terminal velocity**.

On a velocity–time graph this gives a curve that starts steep (gradient = initial acceleration) and flattens off towards a horizontal line at the terminal velocity.

DEFINITION

Terminal velocity is the constant, maximum velocity of an object falling through a fluid, reached when the resultant force on it is zero (for a sphere in a liquid, weight = upthrust + viscous drag).

Terminal velocity of a sphere falling through a liquid

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 At terminal velocity the resultant force is zero, so the upward forces balance the weight:

$$W = U + F$$

2. Write the weight using the density of the sphere. Its volume is $\frac{4}{3}\pi r^3$, so its mass is $\frac{4}{3}\pi r^3 \rho_\mathrm{s}$:

$$W = \tfrac{4}{3}\pi r^3 \rho_\mathrm{s}\, g$$

3. The upthrust is the weight of the liquid displaced, which has the same volume but the liquid's density:

$$U = \tfrac{4}{3}\pi r^3 \rho_\mathrm{f}\, g$$

4. The drag at terminal velocity v_t comes from Stokes' law:

$$F = 6\pi\eta r v_\mathrm{t}$$

5. Substitute all three into $W = U + F$ and collect the two "volume" terms:

$$6\pi\eta r v_\mathrm{t} = \tfrac{4}{3}\pi r^3 g\,(\rho_\mathrm{s} - \rho_\mathrm{f})$$

$\rho_\mathrm{s} - \rho_\mathrm{f}$ appears because the upthrust "cancels out" part of the weight.

6. Divide both sides by $6\pi\eta r$. The π cancels and one power of r cancels, and $\frac{4}{3} \div 6 = \frac{4}{18} = \frac{2}{9}$:

$$\boxed{v_\mathrm{t} = \frac{2r^2 g\,(\rho_\mathrm{s} - \rho_\mathrm{f})}{9\eta}}$$

7. Rearranged for the viscosity, which is what Core Practical 2 uses:

$$\eta = \frac{2r^2 g\,(\rho_\mathrm{s} - \rho_\mathrm{f})}{9 v_\mathrm{t}}$$

The result tells you a lot at a glance:

- $v_\mathrm{t} \propto r^2$: double the radius and the terminal velocity goes up **four** times. The weight (minus upthrust) grows with r^3 but the drag only with r, so big balls win.
- $v_\mathrm{t} \propto 1/\eta$: warm the liquid, its viscosity falls, and balls fall faster.
- If $\rho_\mathrm{s} < \rho_\mathrm{f}$ the formula gives a negative answer: the sphere rises at a steady speed instead, like a bubble in a fizzy drink.

That last point surprises people. NOT IN SPEC The time it takes to get close to terminal velocity is set by $\tau = 2r^2\rho_\mathrm{s}/9\eta$, which for a millimetre-sized ball in glycerol is around a millisecond. So in Core Practical 2 the ball really is at terminal velocity long before the first timing mark, but you should still check, as described below.

WORKED EXAMPLE

A steel ball bearing of radius 1.0 mm falls at a terminal velocity of 1.2 cm s^{-1} through an oil of density 960 kg m^{-3}. The density of steel is 7800 kg m^{-3}. (a) Calculate the weight of the ball and the upthrust on it. (b) Calculate the viscosity of the oil. (c) State one assumption you have made.

Show solution

(a) Volume $V = \frac{4}{3}\pi(1.0 \times 10^{-3})^3 = 4.19 \times 10^{-9}\ \text{m}^3$.

$$W = \rho_\text{s} V g = 7800 \times 4.19 \times 10^{-9} \times 9.81 = 3.21 \times 10^{-4}\ \text{N}$$

$$U = \rho_\text{f} V g = 960 \times 4.19 \times 10^{-9} \times 9.81 = 3.95 \times 10^{-5}\ \text{N}$$

(b) At terminal velocity $F = W - U = 3.21 \times 10^{-4} - 0.39 \times 10^{-4} = 2.81 \times 10^{-4}$ N. From Stokes' law:

$$\eta = \frac{F}{6\pi r v} = \frac{2.81 \times 10^{-4}}{6\pi \times 1.0 \times 10^{-3} \times 0.012} = 1.2\ \text{Pa s}$$

(The formula $\eta = 2r^2 g(\rho_\text{s} - \rho_\text{f})/9v_\text{t}$ gives the same answer in one step.)

(c) Any one of: the flow round the ball is laminar (so Stokes' law applies); the ball is a perfect sphere; the ball had reached terminal velocity; the temperature of the oil was constant; the walls of the container don't affect the motion.

COMMON MISTAKE

Don't write "at terminal velocity, weight = drag". For an object in a liquid the upthrust is a significant part of the balance: weight = upthrust + drag. Only drop the upthrust when the fluid is a gas and the object is much denser than it.

Core Practical 2: viscosity by the falling-ball method

CORE PRACTICAL

Use a falling-ball method to determine the viscosity of a liquid

A tall measuring cylinder is filled with a viscous, transparent liquid such as glycerol or washing-up liquid. Rubber bands round the cylinder mark out timing points. Small steel ball bearings are dropped in and timed between the bands.

1. Measure the **diameter** of each ball with a **micrometer** at several orientations and average; halve it to get r.
2. Find the densities: ρ_s from the mass and volume of a number of balls, and ρ_f by weighing a measured volume of the liquid (or use given values).

3. Place the top band well below the surface, so the ball is at terminal velocity before timing starts. Measure the distance d between bands with a metre rule, with your eye level with each band.
4. Drop the ball gently in the centre of the cylinder (tweezers, or a short tube held just above the surface). Time it between the bands with a stopwatch, or film it beside a ruler and step through the video frame by frame. Then $v = d/t$.
5. Check terminal velocity has been reached: use three equally spaced bands and confirm the two time intervals are the same.
6. Repeat for each ball size, measure the temperature of the liquid with a thermometer, and keep it constant.

Graph: $v_\mathrm{t} = \dfrac{2g(\rho_\mathrm{s} - \rho_\mathrm{f})}{9\eta}\, r^2$ has the form $y = mx$. Plot v against r^2: a straight line through the origin with

$$\text{gradient} = \frac{2g(\rho_\mathrm{s} - \rho_\mathrm{f})}{9\eta} \quad \Rightarrow \quad \eta = \frac{2g(\rho_\mathrm{s} - \rho_\mathrm{f})}{9 \times \text{gradient}}$$

Sources of uncertainty and how to reduce them

- **Reaction time** when starting and stopping the stopwatch: use bands far apart so the times are long, repeat and average, or use video analysis or light gates.
- **Parallax** when judging when the ball passes a band: keep your eye level with the band.
- **Radius**: r is squared, so its percentage uncertainty is doubled. Use a micrometer, not a ruler.
- **Temperature**: η changes a lot with temperature, so measure it and avoid warming the liquid (don't hold the cylinder).
- **Walls**: a ball close to the wall of a narrow tube is slowed down. Use a wide cylinder and drop balls down the middle.
- **Air bubbles** sticking to the ball add upthrust: wet the balls in the liquid before dropping them.

Safety: wipe up spills straight away (glycerol makes the floor slippery), clamp the tall cylinder so it can't topple, and retrieve the balls with a magnet.

WORKED EXAMPLE

In a falling-ball experiment a student plots terminal velocity v against r^2 for steel balls (density 7800 kg m⁻³) falling through a liquid of density 1260 kg m⁻³. The best-fit line passes through the origin and has a gradient of $7.6 \times 10^3\ \mathrm{m^{-1}\ s^{-1}}$. (a) Calculate the viscosity of the liquid. (b) The student's value is higher than the value in a data book. Suggest one reason why.

Show solution

(a) The gradient is $2g(\rho_\mathrm{s} - \rho_\mathrm{f})/9\eta$, so

$$\eta = \frac{2g(\rho_\mathrm{s} - \rho_\mathrm{f})}{9 \times \text{gradient}} = \frac{2 \times 9.81 \times (7800 - 1260)}{9 \times 7.6 \times 10^3} = 1.9\ \mathrm{Pa\ s}$$

Check the units: $\mathrm{m\ s^{-2}} \times \mathrm{kg\ m^{-3}} / (\mathrm{m^{-1}\,s^{-1}}) = \mathrm{kg\ m^{-1}\,s^{-1}} = \mathrm{Pa\ s}$.

(b) The liquid was colder than the data-book temperature (viscosity rises as temperature falls); or the walls of a narrow cylinder slowed the balls, making v too small and η too big.

Quick checks

QUICK CHECK

A block floats in water (density 1000 kg m⁻³) with three-quarters of its volume below the surface. What is the density of the block?

- 250 kg m⁻³
- 750 kg m⁻³
- 1000 kg m⁻³
- 1333 kg m⁻³

For a floating object $V_{\mathrm{sub}}/V = \rho_{\mathrm{obj}}/\rho_{\mathrm{f}}$, so $\rho_{\mathrm{obj}} = 0.75 \times 1000 = 750\ \mathrm{kg\ m^{-3}}$. 1333 comes from dividing the wrong way round; a floating object can't be denser than the liquid.

QUICK CHECK

A steel ball is fully submerged in water and then lowered deeper. What happens to the upthrust on it?

- It increases, because the pressure is greater deeper down.
- It decreases, because there is more water above the ball pushing down.
- It stays the same, because the volume of water displaced is unchanged.
- It is zero, because steel is denser than water.

Upthrust is the weight of fluid displaced, and a fully submerged ball always displaces its own volume. The pressure is bigger on both its top and its bottom, but the *difference* between them stays the same. A sinking object still feels upthrust; it just isn't enough to balance its weight.

QUICK CHECK

Which of these is a correct unit for viscosity?

- N m s^{-1}
- kg m s^{-1}
- kg m^{-1} s^{-1}
- N m^{-1}

$\eta = F/6\pi r v$ has units N/(m × m s^{-1}) = N s m^{-2} = Pa s. Writing N as kg m s^{-2} gives kg m s^{-2} × s × m^{-2} = kg m^{-1} s^{-1}.

QUICK CHECK

Two steel balls fall through the same glycerol. Ball Q has twice the radius of ball P. Stokes' law applies to both. What is the ratio (terminal velocity of Q) / (terminal velocity of P)?

- 2
- 4
- 8
- 1, because all objects fall at the same rate

$v_t \propto r^2$. Doubling r makes the weight minus upthrust 8 times bigger (it goes as r^3), but the drag at a given speed only doubles (it goes as r), so the speed must be 4 times bigger for the forces to balance. "Everything falls at the same rate" is only true with no drag and no upthrust.

QUICK CHECK

The glycerol in a falling-ball experiment warms up from 18 °C to 25 °C during the lesson. What happens to the terminal velocity of a ball bearing?

- It decreases, because warm liquids are more viscous.
- It increases, because the viscosity of the glycerol decreases.
- It stays the same, because the densities have not changed.
- It increases, because the upthrust on the ball decreases a lot.

The viscosity of a liquid falls as its temperature rises, and $v_t \propto 1/\eta$. The density of glycerol does fall very slightly when it warms, but that effect is tiny compared with the large drop in viscosity.

QUICK CHECK

A small sphere is falling through oil at its terminal velocity. Which statement is correct?

- The viscous drag equals the weight of the sphere.
- The resultant force is downwards but too small to notice.
- The viscous drag equals the weight minus the upthrust.
- The upthrust equals the weight of the sphere.

At terminal velocity the resultant force is exactly zero: $W = U + F$, so $F = W - U$. If the upthrust equalled the weight, the sphere would not be falling at all.